

Raina Wu

✉ rwu986@mit.edu

📍 Cambridge, MA

🌐 sleepypandu.github.io

🔗 [sleepypandu](#)

📄 [raina-wu-890758326](#)

Education

B.S. Massachusetts Institute of Technology , Computer Science	Cambridge, MA
- GPA 5.0 / 5.0. Expected May 2027. ML: Deep Learning (G), Reinforcement Learning (G), Computer Vision (G), Introduction to Inference (G), Generative AI / Diffusion Lab, NLP*, TinyML*, Robotic Manipulation* Theory: Design & Analysis of Algorithms, Theory of Computation, Abstract Algebra I & II, Set Theory & Logic, Differential Equations Systems: Computer Systems Engineering, Computation Structures, Intro to C & Assembly, Computer Graphics* (G) = graduate-level; * = in progress	Sept 2023 – May 2027

Experience

Hudson River Trading , Algo Dev Intern (incoming)	New York, NY June 2027 – Aug 2027
Applied Compute , Research Intern (incoming)	Jan 2027 – May 2027
IBM Research , Research Intern — RL for LLM reasoning - Ran ablations and reproductions of existing self-distillation pipelines and uncovered failure modes of OPSD and SDPO. - Ran a prefix-advantage study across Qwen3-1.7B/4B/8B characterizing where privileged-teacher signal localizes in a rollout. Submitted to the NeurIPS 2026 workshop on Transitioning from Pre-Training to Post-Training. - Investigating other uses of distillation — whether privileged information can be used in a helpful, non-cheating way.	June 2026 – Aug 2026
Hudson River Trading , WITTI Wintern — quantitative finance	New York, NY Jan 2026 – Jan 2026
MIT-IBM Watson AI Lab , Researcher — RL for generative models - Investigated RL fine-tuning of flow models for molecule and crystal generation. - Co-author on <i>Tailored primitive initialization is the secret key to reinforcement learning</i> (ACL 2026); built the embedding-similarity pipeline used for the paper's diversity analysis, showing Tailor's warm-start data is more reasoning-diverse than the Standard-CoT and 4-STaR baselines.	Cambridge, MA May 2025 – May 2026
Talkdesk , Software Engineering Intern — multi-agent systems Prototyped agentic user memory for a multi-agent system and a multi-agent evaluation framework for a production RAG pipeline.	June 2025 – Aug 2025
MIT DeCoDE Lab , Researcher — generative AI for design Built the sketch-to-3D retrieval pipeline (2D sketch → nearest 3D car mesh) and benchmarked retrieval models and similarity metrics; co-author on the resulting ASME IDETC-CIE 2025 paper.	Cambridge, MA Jan 2025 – Jan 2025
MIT Media Lab , Researcher — AI security & agents - Built attestation for AI agents negotiating inside GPU-backed trusted execution environments; wrote circom circuits producing zkSNARK proofs that the negotiation ran as specified. - Built tool-using agents (Gmail, Google Calendar, Shopify APIs) and an evaluation harness for multi-agent negotiation.	Cambridge, MA Sept 2024 – May 2025

- Proposed and evaluated an algorithm combining path-based link prediction with influence-maximization heuristics to identify future vital nodes in weighted collaboration networks.
- Introduced RA-2 and quasi-local RA-2, new link-prediction similarity metrics; built the simulation and evaluation pipeline (7 link-prediction metrics $\times$ 12 centrality measures $\times$ 8 selection algorithms, under simple and complex contagion); wrote the initial manuscript.
- Co-author (alphabetical order) on papers in *Physica A* (2026) and on arXiv (2024).

Publications

Tailored primitive initialization is the secret key to reinforcement learning

Aug 2026

Y. Yao, G. Zeng, R. Wu, Y. Zhang, D. Zhao, Z.-W. Hong, C. Gan

doi.org/10.18653/v1/2026.acl-long.1537 (ACL 2026 (Volume 1, Long Papers), pp. 33300–33318)

Influence prediction in collaboration networks: An empirical study on arXiv

2026

M. Lin, L. P. Schaposnik, R. Wu

doi.org/10.1016/j.physa.2026.131451 (Physica A: Statistical Mechanics and its Applications, 689, 131451)

AI agents in engineering design: A multi-agent framework for aesthetic and aerodynamic car design

Aug 2025

M. Elrefaie, J. Qian, R. Wu, Q. Chen, A. Dai, F. Ahmed

doi.org/10.1115/DETC2025-169682 (ASME 2025 IDETC-CIE, Volume 3B, 51st Design Automation Conference)

The social sphere model: Heuristic influence prediction in evolving networks

Feb 2024

M. Lin, L. P. Schaposnik, R. Wu

arxiv.org/abs/2402.03522 (arXiv preprint arXiv:2402.03522)

Honors and Awards

- **Putnam Mathematical Competition** — Top 220 (2025), Top 300 (2026)
- **Math Prize for Girls** — National Top 12 (2023)
- **Mathematical Olympiad Program (MOP)** — national top 60 (2022)
- **USAJMO Winner; USAMO Qualifier**

Teaching

- **Teaching Assistant, 18.404 Theory of Computation** — MIT, Fall 2026
- **Teaching Assistant, Athemath** — 2022–2024
- **Curriculum Developer, AlphaStar Academy** — 2021–2022

Skills

Programming

Python, C++, Java, SQL

ML & Systems

RL for LLMs (PPO, GRPO, OPD, self-distillation), verl, vLLM, LoRA, FSDP, PyTorch, NumPy, multi-node GPU training (LSF), Docker, PostgreSQL, agentic systems, embedding calculations, circom / zkSNARKs

Languages

English (native), Mandarin Chinese (fluent)